TE 10 Mode simulation

A

```
clc;
clear;
close all;
%% Waveguide Parameters
a = 22.86e-3;      % Width (m)
b = 10.16e-3;      % Height (m)
L = 100e-3;        % Length (m)
f = 10e9;          % Operating frequency (Hz)
c = 3e8;           % Speed of light (m/s)
mu0 = 4*pi*1e-7;    % Permeability of free space
eps0 = 1/(mu0*c^2); % Permittivity of free space
omega = 2*pi*f;
%% Theoretical Calculations
fc = c/(2*a);
beta = (2*pi*f/c)*sqrt(1-(fc/f)^2);
lambda_g = 2*pi/beta;
fprintf('TE10 Waveguide Simulation\n');
fprintf('Cutoff frequency = %.4f GHz\n',fc/1e9);
fprintf('Operating frequency = %.2f GHz\n',f/1e9);
fprintf('Propagation constant = %.4f rad/m\n',beta);
fprintf('Guide wavelength = %.4f
mm\n',lambda_g*1000);
%% Spatial Grid
x = linspace(0,a,100);
y = linspace(0,b,50);
z = linspace(0,L,100);
[X,Y] = meshgrid(x,y);
%% TE10 Electric Field (Ez = 0, Ey is dominant)
Ey = sin(pi*X/a);
Ex = zeros(size(X));
figure;
```

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Hx = -(beta/(omega*mu0))*cos(pi*XX/a);
Hz = (pi/(a*omega*mu0))*sin(pi*XX/a);
figure;
subplot(1,2,1);
imagesc(x*1000,z*1000,Hx);
set(gca,'YDir','normal');
colorbar;
xlabel('Width x (mm)');
ylabel('Length z (mm)');
title('H_x Distribution');
subplot(1,2,2);
imagesc(x*1000,z*1000,Hz);
set(gca,'YDir','normal');
colorbar;
xlabel('Width x (mm)');
ylabel('Length z (mm)');
title('H_z Distribution');
%% 3D Electric Field Variation Along Length
[X3,Z3] = meshgrid(x,z);
E3 = sin(pi*X3/a).*cos(beta*Z3);
figure;
surf(X3*1000,Z3*1000,E3,'EdgeColor','none');
colorbar;
xlabel('Width x (mm)');
ylabel('Length z (mm)');
zlabel('Normalized E_y');
title('3D TE_{10} Electric Field');
view(45,30);


%% Simulated Cutoff Frequency Input
```

```
imagesc(x*1000,y*1000,Ey);

set(gca,'YDir','normal');

colorbar;

xlabel('Width x (mm)');

ylabel('Height y (mm)');

title('TE_{10} Electric Field Distribution (E_y)');

axis equal tight;

%% Magnetic Field Components (normalized)

% For TE10: Hx and Hz components

[XX,ZZ] = meshgrid(x,z);
```

```
fc_sim = input('Enter simulated cutoff frequency in GHz: ');

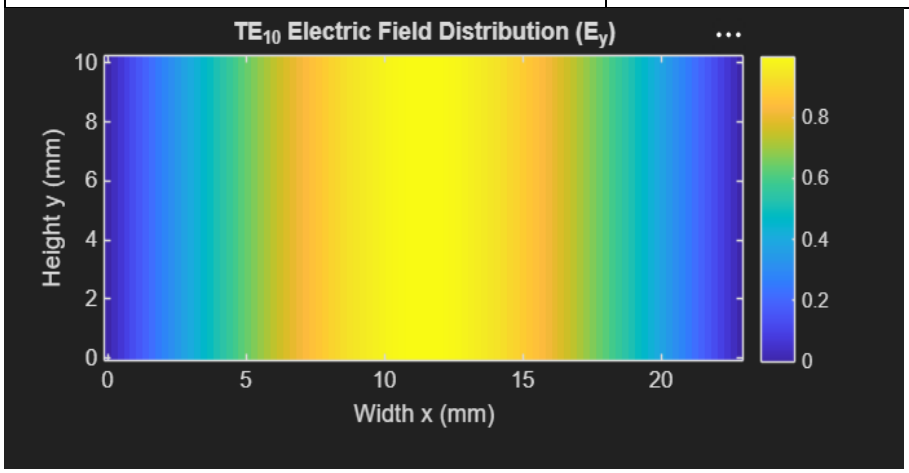
fc_sim = fc_sim*1e9;

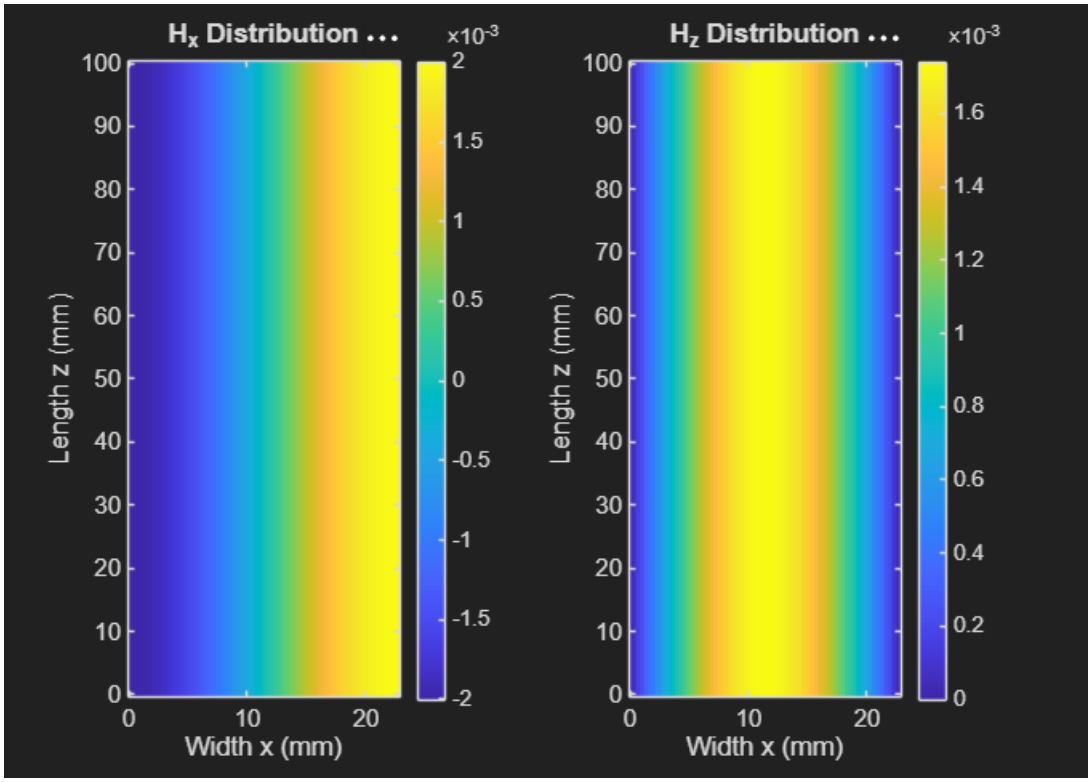

deviation = abs(fc_sim-fc)/fc*100;

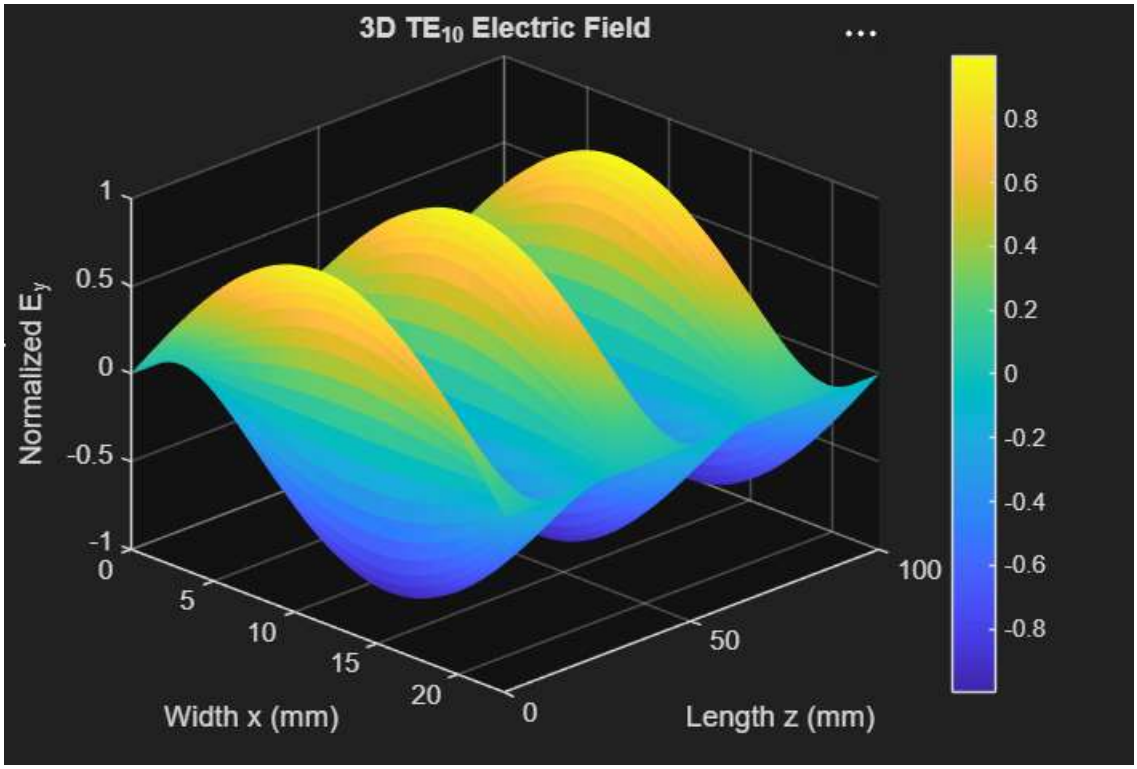
fprintf('Theoretical cutoff frequency = %.4f GHz\n',fc/1e9);

fprintf('Simulated cutoff frequency = %.4f GHz\n',fc_sim/1e9);

fprintf('Percentage deviation = %.4f %%\n',deviation);
```






3D TE10 Electric Field
Normalized Ey
1
0.5
0
-0.5
-1
0
5
10
15
20
Width x (mm)
0
50
100
Length z (mm)
0.8
0.6
0.4
0.2
0
-0.2
-0.4
-0.6
-0.8